

ElboVR: Iterative Development of a VR Application for Post-Surgery Elbow Rehabilitation

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ABSTRACT

Virtual Reality (VR) may help motivate at-home patients to perform the repetitive exercises necessary to regain mobility after orthopedic elbow surgery. This paper details the first three iterations in the development of ElboVR—a VR application developed together with therapists that allow patients to perform elbow exercises and provide physiotherapists with a new way of monitoring patients' Range of Motion (ROM).

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality;

1 INTRODUCTION

Physiotherapy involves repetitive physical exercises and is essential to recovery from orthopedic surgeries. These exercises must be performed several times a day consistently for a specific period to have a positive effect on regaining lost mobility, such as range of motion (ROM) in elbow joints [8]. Daily activities such as cooking, cleaning, eating, or using a computer all depend on the ROM of the patient, making it an important metric for physiotherapists. Physiotherapists are trained in standardized methods to measure their patients' ROM, such as the universal goniometer [2].

Both high-fidelity motion tracking and commercial devices (e.g., the MS Kinect), can be used to track relevant upper-body movements, including ROM, that are relevant to rehabilitation [6]. However, to our knowledge, there are no studies on how commercially available Virtual Reality (VR) can measure ROM based on data from the head-mounted display (HMD) and the controllers alone.

Additionally, the repetitive nature of the exercises and the requirement to consistently perform these exercises several times a day underline the importance of motivation toward physiotherapy treatment. Unfortunately, there is a general lack of motivation and a tendency to stop doing the exercises from home in individuals who need physiotherapy. Slater and Sanchez-Vives [11] have argued that the real power of VR in relation to physiotherapy lies in its ability to improve traditional practices by making them more interesting for patients. A recent systematic review on applications of VR in physiotherapy [5] shows a large increase in work on the topic over the past two decades, and the results indicate that 14% of the 152 papers included in the review related explicitly to orthopedic and traumatology. However, only 2% of the studies described in the systematic review from 2022 used immersive VR, while most of the studies used tools such as the Nintendo Wii [3], Microsoft Kinect [4],

and the Sony Eye Toy [12] as a tool for rehabilitation. In another recent review on the effectiveness of VR in orthopedic rehabilitation, only 1 out of 19 included studies use immersive VR intervention, with the majority of the studies using the Nintendo Wii system [7].

In this paper, we present an immersive VR intervention designed through three iterations in close collaboration with physiotherapists working daily with individual joint injuries. More specifically, we present an application designed to motivate individuals post-elbow surgery to perform exercises required to improve their ROM while providing the physiotherapist with data on ROM improvement of their patients.

2 METHOD

This paper presents an iterative approach toward designing, implementing and evaluating a VR elbow injury training and ROM measurement tool. Each iteration was performed over a three-week span, where each sprint was concluded with an evaluation, involving either domain experts (i.e., physiotherapists) or participants recruited at Anonymous university.

2.1 Iteration I

2.1.1 Aim

The aim of the first iteration was to get initial input from physiotherapists on how to use VR best to help patients with severe elbow injuries. Involving physiotherapists in the process of designing technologies for physiotherapy can ensure that the technology is tailored to meet the specific needs of patients and is effectively able to achieve its intended rehabilitation goals. For this purpose, we consulted two physiotherapists from an anonymous hospital. The two therapists were introduced to the possibilities of VR, and subsequently, they provided input on the most essential requirements for such an application.

2.1.2 Prototype

Based on the feedback from the physiotherapists, two main requirements were set for the VR application:

- The ability to measure the ROM of the elbow
- To motivate the user to perform exercises that require them to perform both flexion and extension of the elbow joint to their maximum ROM

Based on this feedback, a VR application was designed, which enabled users to see their hands, wrists, and arms in a featureless virtual environment (VE), and estimate the users current ROM and display this information in the VE. Moreover, the application included a simple apple picking activity which forced users to perform both elbow extensions.

The Oculus Quest 2 was chosen as the primary device for the intervention. The Quest 2 provides 3D positional data related to the VR controllers and HMD, which served as proxies for the positions of the user's hands and head, respectively. The positions of the user's lower and upper arms were estimated using inverse kinematics,

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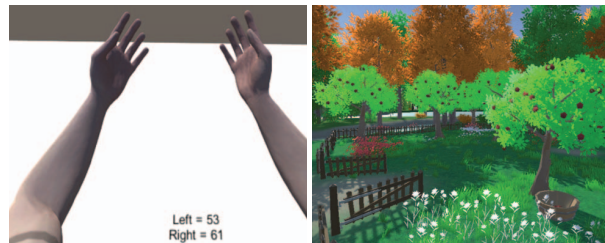


Figure 1: Screenshot of Iteration I showing the user's arms and the ROM for each arm (left) and the VE used for the apple picking exercise (right).

and the ROM was derived from this information. To enable better alignment between the user's real and virtual body, a calibration phase was required. The calibration asked the users to indicate their height as well as the position of their real wrist, which is done by aligning the wrist to a virtual line in the application. Once calibration was complete, the user could see a virtual arm, wrist, and hand aligned with their own hand, wrist, and arm, as well as the ROM of their arms, as seen in Figure 1 (left). Finally, an apple-picking activity was implemented in which the user had to perform an extension of their elbow to pick virtual apples and then flexion of the elbow and place the apples inside a virtual basket (Figure 1, right).

2.1.3 Evaluation

The first iteration was shown to the two physiotherapists working daily with patients suffering from elbow injuries at Anonymous Hospital. Also present was one patient with a severe elbow injury that required surgery. The patient was excited about the application and its future possibilities as he stated that the current exercises at anonymous hospital are boring and repetitive, and he could see the motivational potential of the application. However, he stated that the virtual elbow did not mirror his physical arm and wrist movement perfectly. This can probably be attributed to inaccuracy in the inverse kinematics implementation. Additionally, the patient was able to cheat the system and get a better ROM number by extending his shoulder. After discussions with the user, it was decided that the next iteration should only visualize the hands of the user instead the whole arm since the inconsistency between the virtual representation and the user's actual arm movement can distract the user while the hand position and orientation can be visualized for the user with precision. Moreover, the therapist also recommended only visualizing the hands because she feared that it could demotivate the patient if they could see their arm not fully stretched. Especially since the virtual experience should subconsciously make the patient believe their elbow had more mobility and strength so that they could perform tasks they otherwise thought impossible. Finally, the physiotherapists asked for another motivational mechanism to be implemented for the user to perform flexion and extension of the elbow, such as navigating a 3D space by performing this task. Finally, in regards to the apple-picking activity, the physiotherapists asked for it to be extended so that the user could perform both flexion and extension exercises while also practicing coordination with the healthy arm.

2.2 Iteration II

2.2.1 Aim

The aim of this iteration was to incorporate the feedback received by the physiotherapists and the patient.

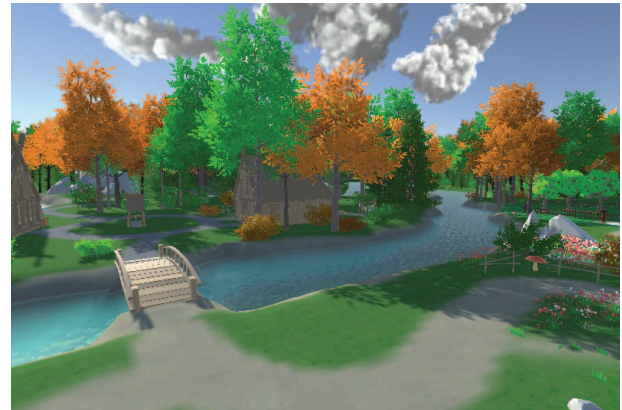


Figure 2: The VE and the apple trees used for the apple-picking activity and exploration in Iteration II

2.2.2 Prototype

In this iteration, only a 3D model of the user's hand was visualized, while the 3D arm and wrist were removed. Illustrating the precise hand position is much easier since the HMD controller provides precise 3D coordinates of the users' hands. A 3D virtual world was also designed to support exploration, as seen in Figure 2.

The player can move around and explore the 3D environment by performing one of the main exercises for individuals with elbow injuries: extending and retracting the arm into their maximum ROM. Therefore, during the calibration phase of the application, the users are now asked to start by demonstrating their maximum and minimum ROM capabilities.

Additionally, the "apple picking" part of the application was changed based on the feedback from the physiotherapists. The user now has to pick apples from a tree by extending the arm, shifting the apple to the other hand, and then placing it into a basket. This activity aims to include arm coordination and using the injured arm in cooperation with the healthy arm.

2.2.3 Evaluation

This iteration of the application was evaluated in Anonymous Laboratory. The same patient who was recovering from severe elbow surgery and one of the physiotherapists visited the lab to help evaluate this iteration of the application.

Once the system was calibrated based on the user's wrist position and height, the participants were asked to navigate to the apple tree using the elbow flexion and extension mechanisms. Then, they were asked to pick ten apples from the tree and place them in their basket.

The test results showed a few flaws that needed fixing. During the calibration, the user had a hard time understanding how to calibrate their wrist position since now their virtual wrist was removed from the application. Additionally, the participants stated that the system did not provide any feedback on how close they were to approach their minimum and maximum ROM when trying to move around in the virtual environment.

This was for them to see how much further, they had to move the arm to take another step, but also to increase the users' motivation. He mentioned that it would not only make them want to push themselves harder, but also, make it more convenient and remove unnecessary confusion by clearly showing how much further they had left before moving.

The physiotherapist stated that the users should be able to save their profile so they do not have to go through the calibration phase each time and also so that in the future, the physiotherapist can

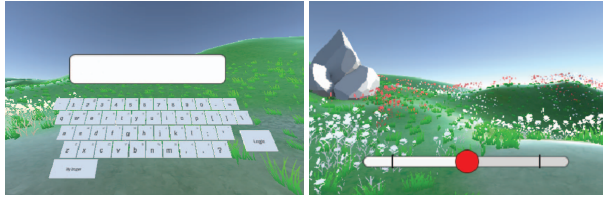


Figure 3: Virtual keyboard used to create an account (left) and slider indicating how far users are from their minimum and maximum arm extension, set during calibration. Movement forward requires alternate movement between these two degrees.

be provided with data regarding the improvement of their patients' ROM using the system.

Finally, the physiotherapist observed that the patient moved their shoulder, resulting in a better ROM Score and smoother movement in the application. This is something that should be restricted, and the physiotherapist asked to create signifiers in the application that would remind the patient not to move their shoulder when performing tasks in the application.

2.3 Iteration III

2.3.1 Aim

The aim of this iteration was to re-design the calibration interaction so that it is easy for the users to use. Additionally, visual signifiers were added so the user can see how far they are from their minimum and maximum ROM during flexion and extension of the elbow, as well as a warning signifier that would activate if they use their shoulders to get better ROM results. Finally, the physiotherapist asked for the ability to track the users' improvement by enabling the users to save their profiles in the application.

2.3.2 Prototype

This iteration initially asks the user to create an account by typing a login name using a virtual keyboard seen in Figure 3(left). This is then saved into a database that will save the users' calibration information as well as their maximum and minimum ROM. Additionally, a user interface was created that explains the steps of the calibration in greater detail. This required a redesign of the interface to be more readable and for instructions to show in a step-by-step-oriented manner. This was done by adding a background to the canvas where the information would be displayed since, previously, the letters could be difficult to spot with the terrain in the background. A slider to indicate the users measured ROM compared to their maximum and minimum ROM is added as seen in figure 3 (right). The X-axis of the slider indicates their minimum and maximum ROM capabilities measured during the calibration phase, while the red dot moves on the slider as the user performs flexion and extensions of the elbow. Finally, a warning signifier is created that will remind the user not to move their shoulder in order to gain better ROM results.

2.3.3 Evaluation

The purpose of this evaluation was twofold: (1) to determine how accurate the application was compared to the goniometer normally used by the therapists to estimate ROM, and (2) to evaluate the perceived usability of the calibration phase and the apple-picking game.

To meet this aim 20 university students were recruited from Anonymous university. None of the participants had known elbow injuries. The study was performed in an open space on campus, and an Oculus Quest 2 and a computer with an RTX 3080 graphics card and a Ryzen 5950X processor were used to run the application. In the first step of the test, the participants were asked to calibrate the

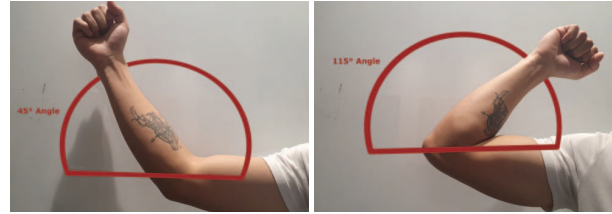


Figure 4: Angles used for the study: 45° (left) and 115° (right)

game so that it know their height and arm length, which formed the basis for the ROM measurements. To simulate an injured elbow, the participants were asked to position their elbow at 45° and 115° (Figure 4). This is an example of a post-elbow surgery maximum ROM as a healthy elbow joint can perform flexion between 130° to 154° and extension between 75° to 85° [13]. Each position would also be measured with a goniometer by a test conductor who is trained to use the tool. These ROM measurements of the VR application were used to compare the application's accuracy to a traditional ROM measurement device (the goniometer). After the calibration, the participants would reposition in the VE to the exercise of apple picking and complete it by picking ten apples.

After exposure to the VR experience, the participants were asked to fill out would then fill out a questionnaire including eight items (Table 1 asking about the perceived usability of the calibration process (Q1-Q4) and the applied picking game (Q5-Q8). The items were adapted from the PSSUQ questionnaire [9], and they were answered on 7-point rating scales, ranging from 1 ('strongly disagree') to 7 ('strongly agree').

Table 1: The eight items used to assess perceived usability of the calibration process and the apple picking game.

Perceived usability of the calibration process

- Q1: The calibration of the steps was confusing
- Q2: I found it intuitive to follow the on-screen instructions
- Q3: I felt frustrated performing the tasks
- Q4: I felt that i could adjust the character to my measurements

Perceived usability of the apple picking game

- Q5: The goal of the task was clear
- Q6: I felt frustrated
- Q7: I thought there was too much inconsistency when performing the task
- Q8: I found the control to be intuitive

2.3.4 Results

When the participants used the prototype to reach a 45° ROM, the goniometer measure yielded a mean angle of $M = 47.95^\circ$ ($SD = 7.05$), and when application was used to reach 115°, the goniometer measure produced a mean angle of 126.20 ($SD = 6.81$). According to physiotherapists, the human error rate is approximately 10-15° when using the goniometer. Thus, the results suggest that the current approach has potential. However, further studies are necessary to provide a better estimate of the VR application's accuracy.

The results related to the perceived usability of the calibration phase can be seen in Figure 5 (left). The distribution of scores suggests that most participants did not find the calibration confusing (Q1) or frustrating (Q3). As indicated by the distribution of scores related to Q2 suggests that it varied how intuitive the participants found the instructions, even though most agreed that it was intuitive. Finally, the participants generally thought that they were able to adjust the character to their measurements (Q4). Turning to the results related to the perceived usability of the apple-picking game, which can be seen in Figure 5 (right). The results indicate that the

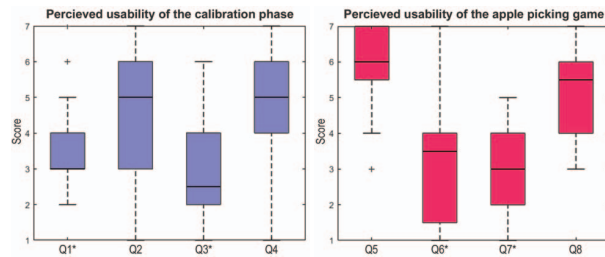


Figure 5: Boxplots showing the results pertaining to perceived usability of the calibration phase (top) and the apple picking game (bottom). Items were answered on 7-point scales, ranging from 1 to 7, and questions highlighted with * used inverted scales (i.e., low scores indicate higher usability)

participants generally found the goal of the task clear (Q5), and most participants also found controls intuitive (Q8). Moreover, most participants did not experience frustration while playing the game (Q6), and they did not experience too much inconsistency when performing the task (Q7). Taken together, we believe these results to indicate that both the calibration phase and the game were perceived as having relatively good usability. However, future studies are needed to get more precise information on the system's usability, including measures of user performance and behavior.

3 DISCUSSION & CONCLUSION

During this project, in cooperation with physiotherapists at Anonymous Hospital, we constructed a VR application with exercises that would help rehabilitation patients with their elbow injuries. The main goal of this study was to explore whether it is feasible to measure the angle of the elbow of the patient using VR HMD and controllers.

To our knowledge, previous studies conducted to apply technology to measure ROM of joints such as the elbow and knee rely on wearable sensors or computer vision technologies.

By calibrating the patients' height and arm length, a calculation could be made to measure the users' ROM on both flexion and extension of the arm. Additionally, signifiers were implemented so that the patients were warned when using their shoulders to "cheat" the system and get a better ROM result.

In general, our results indicate that the VR intervention can be used to calculate the ROM aka. elbow angle with approximately the same error margin as the therapist would measure using a goniometer. The result of the usability evaluation on the calibration shows that to a certain degree, the test participants understood and calibrated the model correctly.

The calibration procedure of the application should be done with the therapist and only be done once. The application instructions are, therefore, only a supplement to how the therapist would guide the patients to calibrate their model correctly. The questionnaire for the apple-picking activity points towards a clear understanding of the task. Further evaluation is required to explore if the application can be used to motivate the patient and support the physiotherapists when working with post-elbow injury rehabilitation. In regards to the patient's motivation, the world needs to include more activities similar to the apple-picking activity to motivate the user further to explore and engage with the application. Additionally, future iterations of the application must look into creating an interface for the psychologists, with which they can have access to their patients' training progress and ROM development which they can use to guide their rehabilitation further.

Technologies such as the VR application proposed in this paper can be used to track the patient's progress and provide real-time feedback to both the patient and the therapist. This information

can be used by physiotherapists to customize the patients' treatment plans based on their specific needs.

Remote physiotherapy services, aka telerehabilitation, can especially benefit patients in remote or rural areas or patients with mobility difficulties. Rehabilitation exercises can be painful for patients recovering from severe injuries. Additionally, VR can be a more cost-effective solution compared to traditional physiotherapy sessions. The patient can use telerehabilitation to save money for traveling to a physical clinic as well as potentially reduce the amount of money needed for the clinician's time. However, according to a review article on the Benefits of Telerehabilitation After Orthopedic Surgeries, the effectiveness is limited if the treatment is taking place via video calls such as Microsoft teams [10].

Pain management in VR has proven efficient and can distract patients from pain and help them relax, reducing their perception of pain and helping them to tolerate physiotherapy exercises [1]. Moreover, physiotherapy sessions with professionals can be expensive and can make it harder for lower-income populations to receive rehabilitation for their injuries.

REFERENCES

- [1] N. Ahmadpour, H. Randall, H. Choksi, A. Gao, C. Vaughan, and P. Poronnik. Virtual reality interventions for acute and chronic pain management. *The international journal of biochemistry & cell biology*, 114:105568, 2019.
- [2] A. D. Armstrong, J. C. MacDermid, S. Chinchalkar, R. S. Stevens, and G. J. King. Reliability of range-of-motion measurement in the elbow and forearm. *Journal of shoulder and elbow surgery*, 7(6):573–580, 1998.
- [3] L. S. Arrebola, L. C. Yi, and V. G. C. de Oliveira. The use of video games combined with conventional physical therapy in children with upper limb fractures: an exploratory study. *Journal of Pediatric Rehabilitation Medicine*, 12(1):65–70, 2019.
- [4] J. M. R. Bacha, G. C. V. Gomes, T. B. de Freitas, L. A. P. Viveiro, K. G. da Silva, G. C. Bueno, E. M. Varise, C. Torriani-Pasin, A. C. Alonso, N. M. S. Luna, et al. Effects of kinect adventures games versus conventional physical therapy on postural control in elderly people: a randomized controlled trial. *Games for health journal*, 7(1):24–36, 2018.
- [5] P. C. A. Brepohl and H. Leite. Virtual reality applied to physiotherapy: a review of current knowledge. *Virtual Reality*, pp. 1–25, 2022.
- [6] C.-Y. Chang, B. Lange, M. Zhang, S. Koenig, P. Requejo, N. Somboon, A. A. Sawchuk, and A. A. Rizzo. Towards pervasive physical rehabilitation using microsoft kinect. In *2012 6th international conference on pervasive computing technologies for healthcare (PervasiveHealth) and workshops*, pp. 159–162. IEEE, 2012.
- [7] M. Gumaa and A. Rehan Youssef. Is virtual reality effective in orthopedic rehabilitation? a systematic review and meta-analysis. *Physical therapy*, 99(10):1304–1325, 2019.
- [8] S. C. Jha, P. Shaky, and P. Baral. Efficacy of physiotherapy in improving the range of motion of elbow after the treatment of pediatric supracondylar humeral fracture. *Birat Journal of Health Sciences*, 3(2):432–436, 2018.
- [9] J. R. Lewis. Psychometric evaluation of the post-study system usability questionnaire: The psuq. In *Proceedings of the human factors society annual meeting*, vol. 36, pp. 1259–1260. Sage Publications Sage CA: Los Angeles, CA, 1992.
- [10] J. M. Pastora-Bernal, R. Martín-Valero, F. J. Barón-López, and M. J. Estebanez-Pérez. Evidence of benefit of telerehabilitation after orthopedic surgery: a systematic review. *Journal of medical Internet research*, 19(4):e6836, 2017.
- [11] M. Slater and M. V. Sanchez-Vives. Enhancing our lives with immersive virtual reality. *Frontiers in Robotics and AI*, 3:74, 2016.
- [12] L. M. Taylor, N. Kerse, T. Frakking, and R. Maddison. Active video games for improving physical performance measures in older people: a meta-analysis. *Journal of geriatric physical therapy* (2001), 41(2):108, 2018.
- [13] E. L. Zwerus, N. W. Willigenburg, V. A. Scholtes, M. P. Somford, D. Eygendaal, and M. P. van den Bekerom. Normative values and

affecting factors for the elbow range of motion. *Shoulder & elbow*, 11(3):215–224, 2019.